

AFFICIONADO COFFEE ROASTERS

Sales Trend and Time-Based Performance Analysis

*A Research Paper on Transaction-Level Exploratory Data Analysis,
Operational Insights, and Data-Driven Recommendations*

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Abstract

This paper presents a transaction-level exploratory data analysis (EDA) of 149,116 point-of-sale records collected in 2025 across three Afficionado Coffee Roasters store locations in New York City — Hell's Kitchen, Astoria, and Lower Manhattan. The objective was to move store operations from intuition-based scheduling toward evidence-based decision-making by quantifying when demand occurs, what drives revenue, and how performance varies across locations and product lines. Following data validation and temporal feature engineering (revenue, hour-of-day, and time-of-day buckets), the analysis establishes that the business generated \$698,812.33 in total revenue from 149,116 transactions at an average order value of \$4.69. Demand is heavily concentrated in the morning: the 6 AM–10 AM "Morning Rush" window alone accounts for 45.8% of total revenue, with 10:00 AM the single busiest hour. Coffee (38.6% of revenue) and Tea (28.1%) are the dominant categories, and the three stores perform within a narrow band of one another (32.9%–33.8% revenue share each), indicating a consistent, replicable business model rather than one location carrying the brand. The paper documents the methodology used, reports the full set of findings produced by the underlying analysis notebook, flags gaps between the original project brief and what the available data could support, and closes with a set of prioritized, evidence-backed operational recommendations.

1. Introduction and Background

In specialty coffee retail, when a sale happens is often as commercially important as what is sold. Staffing, brewing schedules, and inventory replenishment all depend on accurately anticipating the rhythm of customer demand across the day. When this rhythm is not understood quantitatively, retail managers default to intuition — assumptions such as "mornings are busy" — which can lead to overstaffing during genuinely slow hours, understaffing during real rush periods, inconsistent customer experience, and inefficient labor cost allocation.

Afficionado Coffee Roasters operates three stores in New York City and, prior to this project, had transaction-level data but no consolidated, quantitative view of sales trends, peak/off-peak periods, or hourly demand patterns. This project addresses that gap through structured, reproducible time-based sales analytics, with the ultimate goal of supporting staff scheduling, operational planning, and evidence-based business decisions.

2. Problem Statement and Objectives

Despite having granular transaction-level data, the business lacked a consolidated view of sales trends over time, clear identification of peak and off-peak periods, and hourly demand insight broken out by store. As a result, operational decisions were reactive and experience-driven rather than data-driven.

2.1 Primary Objectives

- Identify overall sales trends across the 2025 transaction data.
- Determine the busiest and slowest periods of demand.
- Identify peak transaction hours.

2.2 Secondary Objectives

- Compare temporal demand patterns across the three store locations.
- Support staff scheduling and operational planning with quantitative evidence.
- Provide evidence-based explanations for observed demand patterns.

3. Dataset Description

The dataset consists of 149,116 individual point-of-sale transactions recorded across the three stores in 2025, with 11 original columns per record:

Column	Description
transaction_id	Unique identifier per transaction
year	Transaction year (2025)
transaction_time	Time of transaction (HH:MM:SS)
transaction_qty	Quantity purchased
unit_price	Price per unit (\$)
store_id	Store identifier
store_location	Physical store location
product_id	Unique identifier for each product
product_category	Broad product group (e.g., Coffee, Tea)
product_type	Product variant within the category
product_detail	Detailed attributes such as flavor, blend, or size

Table 1. Original dataset schema (11 columns, 149,116 rows).

It is worth noting for methodological transparency that the dataset contains a time-of-day field (`transaction_time`) and a year field, but no full calendar date field. This has a direct bearing on which parts of the original analytical brief could be executed against this data (see Section 7, Limitations).

4. Methodology

4.1 Data Ingestion and Validation

The raw CSV export was loaded into a pandas DataFrame and validated before any analysis was performed, on the principle that dirty data produces unreliable results. The validation pipeline checked for:

- Correct parsing of `transaction_time` into a proper datetime type (format HH:MM:SS).
- Missing values across all 11 columns.
- Duplicate transaction records.
- Logical consistency: `transaction_qty > 0` and `unit_price > 0` for every row.

The dataset passed validation cleanly: 0 missing values across all columns, 0 duplicate rows, 0 rows with non-positive quantity, and 0 rows with non-positive price. No rows required removal, and the full 149,116-row dataset was carried forward into analysis.

4.2 Feature Engineering

To enable time-based analysis, the following features were derived from the raw fields:

- `revenue = transaction_qty × unit_price` — the monetary value of each transaction.
- `hour` — the hour of day (0–23) extracted from `transaction_time`.
- `minute` — the minute of the transaction, retained for finer-grained analysis if needed.
- `time_period` — a four-way demand bucket applied to each transaction: Morning Rush (7–10), Lunch Rush (12–14), Afternoon (15–18), and Off Peak (all remaining hours, i.e., 6 AM and 7–8 PM in this dataset).

This engineered dataset (15 columns including the five new features) was exported as `afficionado_coffee_cleaned_df_final.csv` and forms the analytical base layer feeding the accompanying Streamlit dashboard.

5. Exploratory Data Analysis and Findings

5.1 Headline Key Performance Indicators

Metric	Value
Total Revenue	\$698,812.33
Total Transactions	149,116
Average Order Value	\$4.69
Best-Performing Store	Hell's Kitchen
Best-Performing Category	Coffee
Best Individual Product	Sustainably Grown Organic (Lg)
Peak Hour	10:00 AM
Peak Time Period	Morning Rush (7–10 AM)

Table 2. Full-year summary KPIs across all three stores.

5.2 Store-Level Performance

Revenue is distributed almost evenly across the three locations, with no single store dominating the business:

Store Location	Revenue	% of Total	Transactions	Avg. Revenue
Hell's Kitchen	\$236,511.17	33.8%	50,735	
Astoria	\$232,243.91	33.2%	50,599	
Lower Manhattan	\$230,057.25	32.9%	47,782	

Table 3. Revenue, transaction volume, and average order value by store.

Hell's Kitchen leads narrowly on both total revenue and transaction count, making it the highest-traffic and highest-earning location. However, the gap between the top and bottom store is under 3% of revenue, which suggests a highly consistent, replicable operating model rather than one location subsidizing weaker ones. Notably, Lower Manhattan has the fewest transactions of the three stores but the highest average revenue per transaction (\$4.81), implying a customer base that buys fewer but higher-value items — a pattern consistent with a denser business-district location where customers may favor premium or bundled purchases (e.g., coffee beans, branded merchandise) over quick single-item stops.

5.3 Product and Category Analysis

Revenue is concentrated in two categories — Coffee and Tea — which together account for two-thirds of all revenue:

Product Category	Revenue	% of Total	Units Sold
Coffee	\$269,952.45	38.6%	89,250
Tea	\$196,405.95	28.1%	69,737
Bakery	\$82,315.64	11.8%	23,214
Drinking Chocolate	\$72,416.00	10.4%	17,457
Coffee Beans	\$40,085.25	5.7%	1,828
Branded	\$13,607.00	1.9%	776
Loose Tea	\$11,213.60	1.6%	1,210
Flavours	\$8,408.80	1.2%	10,511
Packaged Chocolate	\$4,407.64	0.6%	487

Table 4. Revenue and unit volume by product category, ranked by revenue.

At the product-type level, Barista Espresso (\$91,406.20), Brewed Chai Tea (\$77,081.95), and Hot Chocolate (\$72,416.00) are the top three individual revenue drivers. At the most granular level (product_detail), the single top-selling item is "Sustainably Grown Organic (Lg)" at \$21,151.75, closely followed by "Dark Chocolate (Lg)" (\$21,006.00) and "Latte (Rg)" (\$19,112.25) — indicating that large-size, premium/specialty variants outperform standard-size staples even within the same product line.

Price positioning varies meaningfully by category: Coffee Beans command the highest average unit price (\$21.02, range \$10–\$45) as a retail take-home product, followed by Branded merchandise (\$17.72) and Loose Tea (\$9.27). Core beverage categories — Coffee (\$3.02 average) and Tea (\$2.82 average) — are priced as everyday, low-friction purchases, which aligns with their high transaction volume.

5.4 Time-of-Day Demand Analysis

This is the analytical core of the project. Store hours in the dataset run from roughly 6:00 AM to just after 8:00 PM, with transaction volume revealing a clear, single-peaked demand curve rather than the dual morning/evening pattern common in some retail categories:

Hour	Revenue	Transactions
6:00 AM	\$21,900.27	4,594
7:00 AM	\$63,526.47	13,428
8:00 AM	\$82,699.87	17,654
9:00 AM	\$85,169.53	17,764

Hour	Revenue	Transactions
10:00 AM	\$88,673.39	18,545
11:00 AM	\$46,319.14	9,766
12:00 PM	\$40,192.79	8,708
1:00 PM	\$40,367.45	8,714
2:00 PM	\$41,304.74	8,933
3:00 PM	\$41,733.10	8,979
4:00 PM	\$41,122.75	9,093
5:00 PM	\$40,134.31	8,745
6:00 PM	\$34,286.20	7,498
7:00 PM	\$28,446.68	6,092
8:00 PM	\$2,935.64	603

Table 5. Revenue and transaction volume by hour of day, all stores combined.

Demand climbs sharply from 6 AM, peaks at 10:00 AM (18,545 transactions, \$88,673.39 in revenue — the single busiest hour on both measures), then drops by nearly half within the next hour as the late-morning commute window closes. From 11 AM through 6 PM, demand settles into a long, remarkably flat plateau at roughly 8,700–9,100 transactions per hour, before declining steadily through the evening and tapering off sharply after 7 PM. The 8 PM figures represent only a fragment of an hour (stores appear to close shortly after 8:00), rather than a genuine hourly period.

Aggregating into the four time-period buckets makes the concentration of demand explicit:

Time Period	Definition	Revenue	% of Total
Morning Rush	7 AM – 10 AM	\$320,069.26	45.8%
Afternoon	3 PM – 6 PM (+11 AM in txn count)	\$157,276.36	22.5%
Lunch Rush	12 PM – 2 PM	\$121,864.98	17.4%
Off Peak	6 AM, 7 PM – 8 PM, and other hours	\$99,601.73	14.3%

Table 6. Revenue by time-of-day bucket. Morning Rush alone generates nearly half of all daily revenue.

The single most important finding of this project is that Morning Rush (a four-hour window representing roughly a third of the operating day) generates 45.8% of total revenue. Combined with Lunch Rush, just

six hours out of a roughly fourteen-hour operating day account for 63.2% of all revenue — meaning operational readiness during these two windows disproportionately determines business performance.

5.5 Price and Quantity Distribution

The overwhelming majority of transactions are small, quick purchases: 58.5% of transactions involve exactly one item and 39.3% involve two items, together accounting for 97.8% of all transactions. Orders of three or more items are rare (2.2% or less), confirming that Afficionado Coffee Roasters operates a high-frequency, low-basket-size model typical of a coffee counter rather than a grocery or bulk retail format.

A small number of high-value transactions still matter disproportionately: the top 1% of transactions by revenue (1,545 transactions, each worth \$18.00 or more) generated \$41,019.50 — about 5.9% of total revenue from just 1% of transactions. The most common item within this high-value group is the "I Need My Bean!" T-shirt, indicating that branded merchandise, not beverages, drives the largest individual transaction values.

6. Consolidated Key Insights

- Demand is single-peaked and morning-dominated: the 7–10 AM Morning Rush window alone drives 45.8% of daily revenue, with 10:00 AM the single busiest hour by both revenue and transaction count.
- Six hours (Morning Rush + Lunch Rush) generate 63.2% of total revenue — a small operating window carries the majority of the business.
- The three stores perform within a narrow 33.8%–32.9% revenue-share band, indicating a consistent, well-replicated business model rather than dependence on one flagship location.
- Lower Manhattan has the fewest transactions but the highest average order value (\$4.81), suggesting a higher-spend customer mix relative to the other two stores.
- Coffee and Tea together account for 66.7% of total revenue, with Barista Espresso, Brewed Chai Tea, and Hot Chocolate as the leading individual product types.
- Large-size ("Lg") specialty variants outperform standard-size equivalents at the top of the product-detail ranking, pointing to upsell potential in size and premium blends.
- 97.8% of transactions are for one or two items — the core business is high-frequency, low-basket-size, consistent with quick-service coffee retail.
- Branded merchandise (e.g., T-shirts), while a small share of overall revenue (1.9%), disproportionately drives the highest-value individual transactions.

- Demand falls off a cliff after 7 PM, with the 8 PM window contributing under 0.5% of daily revenue — evening hours are a clear over-staffing risk if scheduled at daytime levels.

7. Scope Limitations

For transparency, it should be noted that the underlying dataset contains a year field and a time-of-day field, but no full calendar date (day/month) field. As a result, several secondary objectives set out in the original project brief — daily revenue trends across 2025, weekly aggregation, identification of the busiest and slowest days of the week, and true hour-by-store cross-location heatmaps — could not be executed against this data as delivered, and are not present in the underlying analysis notebook. The completed analysis instead delivers a robust hour-of-day and time-period view, together with store-level and product-level breakdowns, which directly satisfies the primary objectives (overall trends, peak/off-peak identification, peak hour identification) and partially satisfies the secondary objective of cross-store comparison (via aggregate store totals, though not hour-by-hour store comparison).

Should day-level granularity become available in a future data extract (e.g., a full `transaction_date` field), the same feature-engineering approach used here (Section 4.2) can be directly extended to compute day-of-week aggregates, weekly trend lines, and per-store hourly heatmaps, closing this gap without requiring a change in methodology.

8. Business Recommendations

8.1 Staffing and Scheduling

- Concentrate peak staffing between 7:00 AM and 10:00 AM across all three stores — this window alone drives nearly half of daily revenue and should be treated as the non-negotiable core shift.
- Maintain a secondary, moderate staffing tier through the 11 AM–6 PM plateau, where demand is flat but steady (roughly 8,700–9,100 transactions/hour) rather than declining.
- Taper staffing sharply after 7 PM and minimize coverage after 8 PM, where demand drops to a small fraction of daytime levels.

8.2 Merchandising and Product Strategy

- Prioritize Coffee and Tea inventory and prep capacity, as they jointly drive two-thirds of revenue.
- Promote large-size ("Lg") and specialty/organic variants at the point of sale, since these already outperform standard sizes and represent a clear upsell lever.

- Feature branded merchandise (e.g., T-shirts) near checkout, given its outsized role in driving high-value transactions despite low overall volume.

8.3 Store-Level Actions

- Use Hell's Kitchen's marginally higher traffic and revenue as a template for staffing ratios and product mix at the other two stores, while monitoring whether Lower Manhattan's higher average order value reflects an opportunity to test premium offerings at the other locations.

8.4 Off-Peak Demand Building

- Run targeted promotions (e.g., loyalty bonuses, bundle pricing) during the Off-Peak and late-Afternoon windows to smooth the demand curve and improve utilization of fixed labor and space costs during otherwise slow hours.

8.5 Data and Analytics Roadmap

- Capture a full transaction date field (not just year) in future data extracts to unlock day-of-week and weekly trend analysis, and true store-by-hour heatmaps, as originally scoped.
- Operationalize the findings in the accompanying Streamlit dashboard (store filter, day-of-week selector, hour-range slider, revenue/quantity toggle) so store managers can self-serve these insights on an ongoing basis rather than relying on static reports.

9. Conclusion

This analysis converts 149,116 raw point-of-sale records into a clear, quantitative picture of when and how Afficionado Coffee Roasters earns its revenue. The data confirms the intuitive belief that mornings are busy, but sharpens it into an actionable fact: nearly half of all revenue is earned in a single four-hour morning window, and 10:00 AM specifically is the busiest hour across the business. Combined with a well-balanced three-store footprint and a clear category and product hierarchy, these findings give the business a concrete, evidence-based foundation for staff scheduling, inventory planning, and promotional strategy — directly addressing the reactive, intuition-driven decision-making identified in the original problem statement. Closing the remaining data gap around full calendar dates would allow the same methodology to extend naturally into day-of-week and weekly trend analysis in a future iteration of this project.